Care Transition Efficiency & Placement Outcome Analytics

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Abstract

This research analyzes the efficiency of children transitioning through the CBP to HHS care pipeline and eventual sponsor placement. The objective is to evaluate transfer efficiency, discharge effectiveness, backlog formation, and overall pipeline throughput. Using derived KPIs and temporal analysis, the study identifies system bottlenecks and opportunities for improving reunification timelines.

Introduction

The Unaccompanied Alien Children (UAC) program functions as a multi-stage care pipeline involving apprehension, CBP custody, transfer to HHS care, and sponsor placement. Monitoring only aggregate counts does not reveal process efficiency. This study reframes the dataset to evaluate operational performance and identify delays.

Problem Statement

While aggregate counts of children in custody are monitored, process efficiency metrics are largely absent. Key unanswered questions include how efficiently children are transferred, whether discharges keep pace with inflows, where backlogs accumulate, and whether placement outcomes improve over time.

Project Objectives

Measure CBP to HHS transition efficiency, evaluate discharge outcomes, and identify process bottlenecks. Secondary objectives include supporting faster reunification, improving workflows, and informing policy reforms.

Dataset Description

The dataset includes Date, Children apprehended and placed in CBP custody, Children in CBP custody, Children transferred out of CBP custody, Children in HHS Care, and Children discharged from HHS Care.

Methodology

The pipeline is modeled as CBP custody to HHS care to sponsor placement. Metrics include Transfer Efficiency, Discharge Effectiveness, and Throughput. Backlog is identified using inflow vs exit imbalance.

KPIs

Transfer Efficiency Ratio, Discharge Effectiveness Index, Pipeline Throughput, Backlog Accumulation Rate, Outcome Stability Score.

Analysis & Insights

Transfer efficiency fluctuates, discharge effectiveness is lower than transfer rate, backlog accumulates during high inflow, and placement stage acts as bottleneck.

Conclusion

The project identifies inefficiencies in the care transition pipeline and highlights need for improved placement workflows to reduce backlog and improve reunification timelines.